

Structural-Lexical Friction: A Cross-Register Computational Study of Text Difficulty

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Abstract

Classical readability formulas predict text difficulty from surface counts and bring grammatical structure into the calculation only through sentence length. This paper introduces *structural-lexical friction* — the cognitive cost a reader pays when lexical and syntactic demands rise together — and measures it with a transparent index set across 16,904 English passages. The corpus pulls from four high-density informational registers (anchored by the Korean College Scholastic Ability Test and LogiQA) and five baseline registers. A pre-pooling Mann-Whitney U test shows that CSAT and LogiQA cannot be combined: every parser-derived index distinguishes them. The headline ablation runs in two stages — length-controlled classification, then a length-matched sub-corpus restricted to 150–190 words per passage. Parser-aware features yield a reliable AUC gain over surface-only baselines ($p < 10^{-4}$, paired DeLong), and the gap grows as length is held first by feature exclusion and then by sample selection. A word-count-matched adversarial design then separates the construct cleanly. Mean Clausal Depth (*MCD*) shifts by 1.36 units ($p < 10^{-4}$) while Flesch Reading Ease registers no detectable movement. Three independent dependency parsers reproduce the headline signal, ruling out a single-pipeline artefact. What looks like redundancy of parser-aware indices in conventional readability tasks turns out to be mechanical, since Flesch's *W/S* term absorbs clausal embedding through sentence length; once length is pinned down physically, the structural signal recovers in full.

1. Introduction

Reading under time pressure is a problem of constrained resource allocation: working memory and attention are finite, and the reader must spend both on every lexical, syntactic, and discourse decision the text supplies. A constituent that fails to integrate before the budget runs out drags down whatever comes after it. The consequences become measurable wherever misreading carries a cost — admissions tests, legal instruments, technical specifications, clinical documents — where item-level accuracy, response latency, and downstream task performance all respond to text difficulty in ways a practical theory has to predict rather than merely describe.

The dominant family of text-difficulty instruments was engineered for that practical purpose. Readability formulas in the tradition of Flesch (1948), reviewed in Klare (1974), predict difficulty from cheap surface counts: average sentence length, average syllables per word, sometimes a word-familiarity check against a frequency list. They have lasted because those surface features genuinely carry signal, but that same cheapness becomes a ceiling once questions of structure enter the picture. Two passages with identical Flesch Reading Ease can differ sharply in how deep their subordinate clauses recurse, in how dense their referential chains run, and in how much working memory they cost to track, and a formula blind to those features cannot rank such passages against one another.

Our aim is not to replace classical readability metrics but to characterise empirically, across a multi-register corpus, how surface and structural indices actually relate to one another in practice. The joint cost we call *structural-lexical friction*: the load a reader pays for assembling a coherent representation when unfamiliar vocabulary, deep clausal embedding, and dense referential structure all pull at the same time. We chose this descriptor over more philosophically loaded labels such as "epistemic friction" because epistemological vocabulary belongs to accounts of knowledge and belief, not to computational measurements of grammatical and lexical complexity.

The paper contributes on three fronts: corpus design with diagnostics, the primary classification experiments, and robustness validation. The corpus is a 16,904-passage cross-register assembly. Four high-density informational registers — CSAT reading-comprehension passages, LogiQA logical-reasoning passages, *arXiv* abstracts, and U.S. federal judicial opinions with inline reporter and statutory citations stripped out — sit against five baseline registers: Brown Corpus, Project Gutenberg fiction, *WikiText-103*, Reuters newswire, and OneStopEnglish at the ELE and INT levels. Before any analysis we put CSAT and LogiQA through a Mann-Whitney U *pre-pooling* check, which finds that every parser-derived index splits them systematically; the two therefore enter the analysis as separate strata rather than as a combined assessment register.

The experimental programme runs four classification designs. The first is a full-feature between-register classification, with paired DeLong AUC comparisons setting the baseline discrimination. The second drops length-coupled features and re-runs the classification, which isolates what dedicated syntactic indices contribute once the lexical-length signal of *FRE* is taken away. The third uses a length-matched sub-corpus in which every passage is physically constrained to 150–190 words, removing residual register-typical-length confounds. The fourth is intra-register: inside each of four baseline registers, the lowest-*FRE* quartile is distinguished from the highest-*FRE* quartile under a feature set with *FRE* itself excluded. The fourth design asks whether the syntactic indices carry difficulty signal *inside* a domain, not just across the high-vs-baseline split.

A word-count-matched adversarial transformation supplements the four classification designs and replaces an earlier stage-wise embedding design that turned out to be mathematically circular. The headline clausal-depth metric is then put through a three-way parser robustness pilot (spaCy small, spaCy transformer, Stanford *Stanza*), a full-corpus re-parse with the transformer model on an earlier corpus version, and a 500-passage *Stanza* re-parse on the v3 subset. All classifications are re-run under nested 5-fold cross-validation with inner-loop random hyperparameter search.

One framing note: this paper sits in the methodological-paper tradition as an *analytical tool validation study* rather than as a primary psycholinguistic investigation. No behavioural data is collected here, so no claims follow about reading times, fixation patterns, or comprehension accuracy in any specific population. *Structural-lexical friction* serves as a placeholder for the load that cognitive accounts document elsewhere. Our job is to characterise the *structural signatures* those accounts predict, to identify which existing indices best instrument them, and to lay analytic groundwork that later iterations can cross-reference against public eye-tracking and self-paced-reading data — Dundee (Kennedy & Pynte, 2005), GECO (Cop, Dirix, Drieghe, & Duyck, 2017), Provo (Luke & Christianson, 2018).

2. Literature Review

The cognitive-architectural case for treating reading as load-bearing was made early. Just and Carpenter (1992) put forward a capacity theory of comprehension on which working-memory span propagates directly into reading-time and comprehension differences across individuals. Working memory is, in their framework, a limited pool of activation shared across syntactic parsing, semantic integration, and inferential elaboration. Texts that demand high simultaneous activation overload the reader regardless of how rare any single word may be. Sweller's (1988) cognitive load theory, developed for problem-solving instruction, supplies the matching vocabulary: intrinsic load (the irreducible complexity of the material), extraneous load (cost imposed by the presentation), and germane load (the productive cost of schema construction). Classical readability formulas reach only the second of these.

Work on syntactic processing has long isolated the structural features that carry the heaviest load. Gibson (1998) put forward Dependency Locality Theory, on which the integration cost of a syntactic dependency rises with the number of intervening discourse referents. The predicted high-cost configurations are embedded relative clauses, garden paths, and stacked clausal complements. Eye-movement evidence collected over two decades and synthesised by Rayner

(1998) backs this up in fine grain: regression rates, fixation durations, and word-skipping probabilities all track properties that classical readability scores cannot see. At the discourse level Kintsch's (1988) Construction-Integration model frames comprehension as a two-stage settling of a propositional network. Downstream integration succeeds only when the activated network is locally coherent. McNamara, Kintsch, Songer, and Kintsch (1996) showed empirically that this coherence requirement interacts with prior knowledge in non-obvious ways.

A second, broadly computational strand has tried to operationalise these constructs over large text collections. Graesser, McNamara, Louwerse, and Cai's (2004) Coh-Metrix system computes more than a hundred indices spanning lexical, syntactic, referential, and situational levels. Lu (2010) built a dedicated parser-based measure of syntactic complexity tailored to second-language writing. Kyle and Crossley (2018) extended the L2-syntactic toolbox with fine-grained usage-based indices. Arc-distance indices in this family — mean arc length, arc-crossing depth — are known to be sensitive to specific attachment decisions made by individual parsers, especially for coordinate structures and prepositional-phrase attachment; cross-parser agreement is quantified explicitly in §4.6. Crossley, Skalicky, and Dascalu (2019) show that combining diverse indices in statistical models recovers human readability judgments more accurately than any single surface formula. Biber, Larsson, and Hancock (2024) apply structural-equation modelling to four theory-based models of grammatical complexity and find that the construct decomposes into structurally distinct families rather than reducing to a single dimension. Vajjala and Lucic (2018) released the OneStopEnglish parallel corpus, where the same underlying article exists at three reading levels, as a controlled testbed. Liu, Cui, Liu, Liu, Wang, and Zhang (2020) released LogiQA, a public corpus of expert-level logical-reasoning passages translated from China's national civil-service examinations, which we use here as a complement to CSAT. On the inferential side, Barr, Levy, Scheepers, and Tily (2013) raised the bar for confirmatory hypothesis testing on multi-item, multi-participant data in reading research.

What this literature has not done is settle on what these multi-index efforts are actually *measuring*. The terms "complexity", "cohesion", "load", and "difficulty" appear interchangeably, without always naming the cognitive bottleneck each one approximates. *Structural-lexical friction* is a deliberately narrow placeholder: the load that arises when lexical and syntactic demands rise together. It is not so much a new metric as a frame for relating existing ones, and one empirical task of this paper is to show that the relations among those existing indices are subtler than the contrast "surface formula versus parser-based index" lets on.

3. Methods and Mathematical Framework

3.1 Corpus

The empirical corpus contains 16,904 English-language passages partitioned into two buckets, balanced at roughly a 1.67 : 1 ratio of high-density to baseline (high_density = 10,565; baseline = 6,339).

The *high-density informational* bucket contains four registers. (i) **Korean CSAT reading-comprehension passages** (n = 124), taken from officially administered CSAT items released by the Korea Institute of Curriculum and Evaluation. Because the item form modifies the source passage to fit a target question type (gaps, sentence-order disturbances, sentence-insertion shifts), we work from each item's source passage in its original continuous form, reverting these item-level modifications before metric computation. (ii) **LogiQA reasoning passages** (n = 4,679; Liu et al., 2020), drawn from the public benchmark dataset whose contexts are translated from Chinese civil-service-examination logical-reasoning items. We considered pooling CSAT and LogiQA into a single "high-stakes assessment register" and discarded the pool: a two-sided Mann-Whitney U analysis (§4.1) revealed considerable distributional divergence on *MCD*, *MaxCD*, *DC/C*, *LeftBR*, and *MTLD* (rank-biserial $|r_{rb}| \geq 0.30$, $p < 10^{-7}$); CSAT and LogiQA enter the analysis as distinct registers throughout. (iii) **arXiv research abstracts** (n = 779), fetched live from the *arXiv* API across the categories `cs.CL`, `q-bio`, `physics`, `math`, and

econ and filtered to ASCII abstracts of at least 50 words. (iv) **U.S. federal judicial opinions** (n = 4,999), retrieved from the public CourtListener S3 bulk archive (snapshot `opinions-2026-03-31.csv.bz2`) and processed with a sentence-aligned multi-chunk algorithm that drops case captions and procedural headers heuristically, segments the body into sentences, and accumulates non-overlapping chunks of approximately 320–460 words. Prior to index computation we programmatically strip inline reporter citations (e.g., *410 F.3d 102*), statutory references (e.g., *28 U.S.C. § 1291*), Federal Rules citations (e.g., *Fed. R. Civ. P. 12(b)(6)*), pinpoint citations, and Latin cross-references (*Id.*, *Ibid.*, *supra*, *infra*); the median chunk shortens from 432 to 409 words under this cleanup. Up to three chunks are retained per opinion. The 4,999 chunks were drawn from 2,171 distinct opinions.

The *baseline informational* bucket contains five registers. (i) **Brown Corpus** prose paragraphs (n = 400) sampled from the news, editorial, reviews, lore, learned, religion, and hobbies categories. (ii) **Project Gutenberg** modern public-domain fiction paragraphs (n = 1,849) chunked from eighteen canonical nineteenth- and early-twentieth-century novels. (iii) **WikiText-103** (n = 2,000), drawn from the peer-reviewed benchmark dataset released by Merity, Xiong, Bradbury, and Socher (2016, arXiv:1609.07843), the principal modern-encyclopedic anchor of the design. (iv) **Reuters-21578 newswire** (n = 1,477), the standard NLTK distribution of 1987 Reuters newswire articles. (v) **OneStopEnglish ELE and INT** levels (n = 613, after sentence-aligned rechunking; 266 ELE and 347 INT chunks), the peer-reviewed parallel-corpus release of Vajjala and Lucic (2018). The third release level of OneStopEnglish (Adv) is excluded to keep the baseline cleanly accessible. All sources are public domain, CC-licensed, or open-access.

3.2 Indices

Seventeen per-passage indices are computed and arrange themselves into three families. We use italic abbreviations in prose, and the upright mathematical style (FRE, MCD) inside displayed equations.

Surface (5 indices). Word Count (WC ; `textstat`), Flesch Reading Ease (FRE ; Flesch, 1948), Automated Readability Index (ARI), Type-Token Ratio (TTR) on lower-cased alphabetic tokens, and Mean Sentence Length (MLS). The Flesch formula

$$FRE = 206.835 - 1.015 \left(\frac{W}{S} \right) - 84.6 \left(\frac{\Sigma}{W} \right),$$

with W words, S sentences, and Σ syllables, enters the design as the natural baseline against which any additional index must justify itself.

Lexical-diversity (2 indices). Measure of Textual Lexical Diversity ($MTLD$; McCarthy, 2005) and a nominalisation ratio ($NomR$).

Syntactic (10 indices).

- *Clausal-depth family.* Mean Clausal Depth (MCD) and Maximum Clausal Depth ($MaxCD$); the ratio of complex T-units to T-units (CT/T); the ratio of dependent clauses to clauses (DC/C); Mean Clause Length (MLC). These follow Lu's (2010) tradition.
- *Arc-based family (Gibson-aligned proxies; Gibson, 1998).* Mean Arc Length (MAL), mean per-sentence Maximum Arc Length ($MaxAL$), Left-Branching ratio ($LeftBR$), Mean Arc-Crossing index ($MeanArcCross$), per-sentence Maximum Arc-Crossing ($MaxArcCross$). The arc-based family carries known sensitivity to specific attachment decisions; cross-parser agreement is quantified explicitly in §4.6.

MCD is defined as follows. Each text is parsed with *spaCy* (`en_core_web_sm`; Honnibal & Montani, 2017). For each sentence root, a depth-first traversal counts the maximum number of edges on a root-to-leaf path whose dependency label lies in

$$\mathcal{C} = \text{ccomp, xcomp, advcl, relcl, acl, csubj.}$$

Phrasal-level dependencies (`nsubj`, `dobj`, `prep`, etc.) do *not* increment the counter, so MCD isolates clausal recursion from mere phrasal length. The per-sentence maxima are averaged over

the sentences:

$$\text{MCD} = \frac{1}{|S|} \sum_{s \in S} \max_{p \in \text{paths}(s)} |e \in p : \text{dep}(e) \in \mathcal{C}|.$$

3.3 Between-register classification (full corpus, length-controlled, and length-matched)

We label the 16,904 passages $y = 1$ for the high-density bucket and $y = 0$ for the baseline. Three feature tiers enter the ablation: **T1 — Surface:** $\{FRE, ARI, WC, TTR, MLS\}$; **T2 — + Lexical:** $T1 \cup \{MTLD, NomR\}$; **T3 — Full:** $T2 \cup \{MCD, MaxCD, DC/C, CT/T, MLC, MAL, MaxAL, LeftBR, MeanArcCross, MaxArcCross\}$. A length-controlled supplementary block re-runs the same three classifiers on tiers from which WC and MLS are removed. The length-matched sub-corpus additionally constrains every passage to $[150, 190]$ words by sentence-aligned re-chunking, the narrow band defined by the CSAT distribution.

Three classifiers train under 5-fold cross-validation: L_2 -regularised logistic regression (LR), Random Forest (RF ; 300 trees, balanced class weights), and Gradient Boosting (GBM ; 200 trees, depth 3, learning rate 0.08 in the primary block). Discrimination is reported as cross-validated AUC with 95% bootstrap confidence intervals (500 resamples) and paired-sample DeLong tests. A nested 5-fold cross-validation with inner-loop random hyperparameter search is reported in §4.10 for GBM , the leading classifier in the primary block.

3.4 Intra-register difficulty-quartile classification

To verify that the parser-aware indices carry difficulty signal within a single domain — not only the bucket-distinguishing signal of the headline experiment — we partition four baseline registers (*WikiText-103*, Brown Corpus, Project Gutenberg, Reuters) into difficulty quartiles by their internal FRE distribution. The lowest- FRE quartile (hardest, $y = 1$) is contrasted with the highest- FRE quartile (easiest, $y = 0$). FRE itself is excluded from the feature set to keep the task non-circular.

3.5 Multi-class register classification

To assess whether the indices discriminate individual registers rather than merely the broad binary bucket, one-vs-rest *LR* and *RF* classifiers predict the nine-way register label; macro-averaged and per-class AUC are reported.

3.6 Word-count-matched adversarial transformation

A separate experiment isolates structural complexity from word-count variation. We construct twenty-five sentence pairs, each matched on word count ($\Delta = 0-1$) and on overlapping content vocabulary but contrasted on syntactic architecture: the *low-friction* member uses two coordinated independent clauses ("X did Y and Z did W"), while the *high-friction* member uses a deeply nested complement clause ("That X did Y compelled Z to do W"). When word count is held constant, *FRE*'s *W/S* term cannot move, and only the syllables-per-word term remains free to drift, so any movement in *MCD* between pair members must trace back to syntactic architecture alone. Paired Wilcoxon signed-rank tests are reported across the pair set. The redesign replaces an earlier stage-wise embedding experiment in which each successive depth stage added a reporting clause and the *FRE* drop followed mechanically from *W/S* rising; in its place we test directly whether the parser-aware indices respond to structure once length is physically pinned.

3.7 Parser robustness

To test the sensitivity of the parser-derived indices to the choice of dependency parser, a stratified 50-passage three-way pilot routes the same texts through *spaCy* `en_core_web_sm`, *spaCy* `en_core_web_trf`, and Stanford *Stanza*. An earlier version of the corpus ($n = 12,242$, before the LogiQA expansion and before the citation-cleaning of the legal sub-corpus) was then re-parsed in full with `en_core_web_trf`. For the present v3 corpus, *Stanza* re-parses a stratified 500-passage subset to confirm that the headline *MCD* signal carries over to the LogiQA and citation-cleaned legal registers.

3.8 Multicollinearity diagnostics and PCA compression

The 17 indices submit to a Variance Inflation Factor analysis to identify multicollinear features, and the 10 syntactic indices project additionally through Principal Component Analysis to provide an orthogonal compression of the parser-aware family. Both diagnostics are reported as supplementary in §4.11.

4. Empirical Results

4.1 Corpus-level summary and pre-pooling test

Per-register descriptive statistics appear in Table 1.

Table 1. Per-register passage counts and mean values of selected indices computed with the `en_core_web_sm` primary parser. $N = 16,904$. The arXiv per-register count (763) is below the corpus jsonl count (779) because sixteen abstracts contained non-ASCII tokens or formatting that the parser pipeline rejected at the parse stage; all classification analyses operate on the 16,904 successfully parsed passages.

Bucket	Register	n	Mean <i>WC</i>	Mean <i>FRE</i>	Mean <i>TTR</i>	Mean <i>MCD</i>
high_density	judicial_opinion (cleaned)	4,999	404.1	43.1	0.478	1.091
high_density	logiqa	4,679	93.3	39.4	0.659	1.074
high_density	arxiv_abstract	763	175.1	9.9	0.691	1.122
high_density	csat	124	157.9	39.1	0.642	1.325
baseline	wikitext_modern_encyclopedic	2,000	192.4	48.7	0.612	1.048
baseline	gutenberg_fiction	1,849	198.1	58.2	0.623	1.501
baseline	reuters_newswire	1,477	248.9	51.9	0.554	1.488
baseline	brown_news_general	400	179.7	46.0	0.628	1.140
baseline	onestop_int	347	354.4	57.5	0.560	1.246
baseline	onestop_ele	266	338.7	65.4	0.528	1.085

The pre-pooling Mann-Whitney U analysis of CSAT against LogiQA — a check on whether the two assessment registers could combine into a single high-stakes academic subspace — found

clear divergence on every parser-derived index of interest: *MCD* ($r_{rb} = -0.34$, $p = 8.7 \times 10^{-11}$), *MaxCD* ($r_{rb} = -0.42$, $p < 10^{-16}$), *DC/C* ($r_{rb} = -0.30$, $p = 1.5 \times 10^{-8}$), *LeftBR* ($r_{rb} = +0.35$, $p = 2.9 \times 10^{-11}$), and *MTLD* ($r_{rb} = -0.56$, $p < 10^{-25}$). CSAT shows deeper structural embedding and broader lexical diversity than LogiQA. Pooling would mask exactly these systematic differences. The two registers therefore enter the analysis as distinct strata throughout.

Table 1 needs a direct reconciliation. Two baseline registers — Project Gutenberg fiction ($MCD = 1.501$) and Reuters newswire ($MCD = 1.488$) — show *higher* mean clausal depths than any register in the high-density bucket (judicial-opinion 1.091, *arXiv* 1.122, LogiQA 1.074, CSAT 1.325). The pattern reflects how different registers reach density through different mechanisms. Nineteenth- and early-twentieth-century literary fiction is generically committed to long periodic sentences with stacked subordinate clauses; Dickens, Austen, Hawthorne, Brontë, and Wilde do this routinely. Reuters 1987 newswire nests embedded attributions ("officials said that traders expected the rate to rise") and conditional clauses about market reaction. Both register a high clausal-recursion signature without high informational density. The high-density registers achieve density differently. *arXiv* abstracts compress information into noun-phrase chains and nominalised constructions (*MTLD* 87, *NomR* high), not clausal nesting. LogiQA's translated logical-reasoning contexts decompose argument premises into short propositional sentences ("Some X. All Y. Therefore Z."). The cleaned judicial sub-corpus interleaves short procedural sentences with denser statutory analysis, which pulls the mean down. CSAT, engineered for the high-stakes assessment of L2 readers, is curated to be syntactically demanding ($MCD = 1.325$, the highest in the high-density bucket) without becoming lexically inaccessible. The implication is clear: *MCD* indexes one specific dimension of difficulty, namely clausal recursion. It has to be read alongside nominalisation and lexical-diversity indices that capture the others. No single index — classical or parser-aware — can substitute for the rest, and that is precisely why the classification ablation in §4.2–§4.4 puts the full tier above every restricted tier across every block. The pattern lines up with Biber, Larsson, and Hancock's (2024) structural-

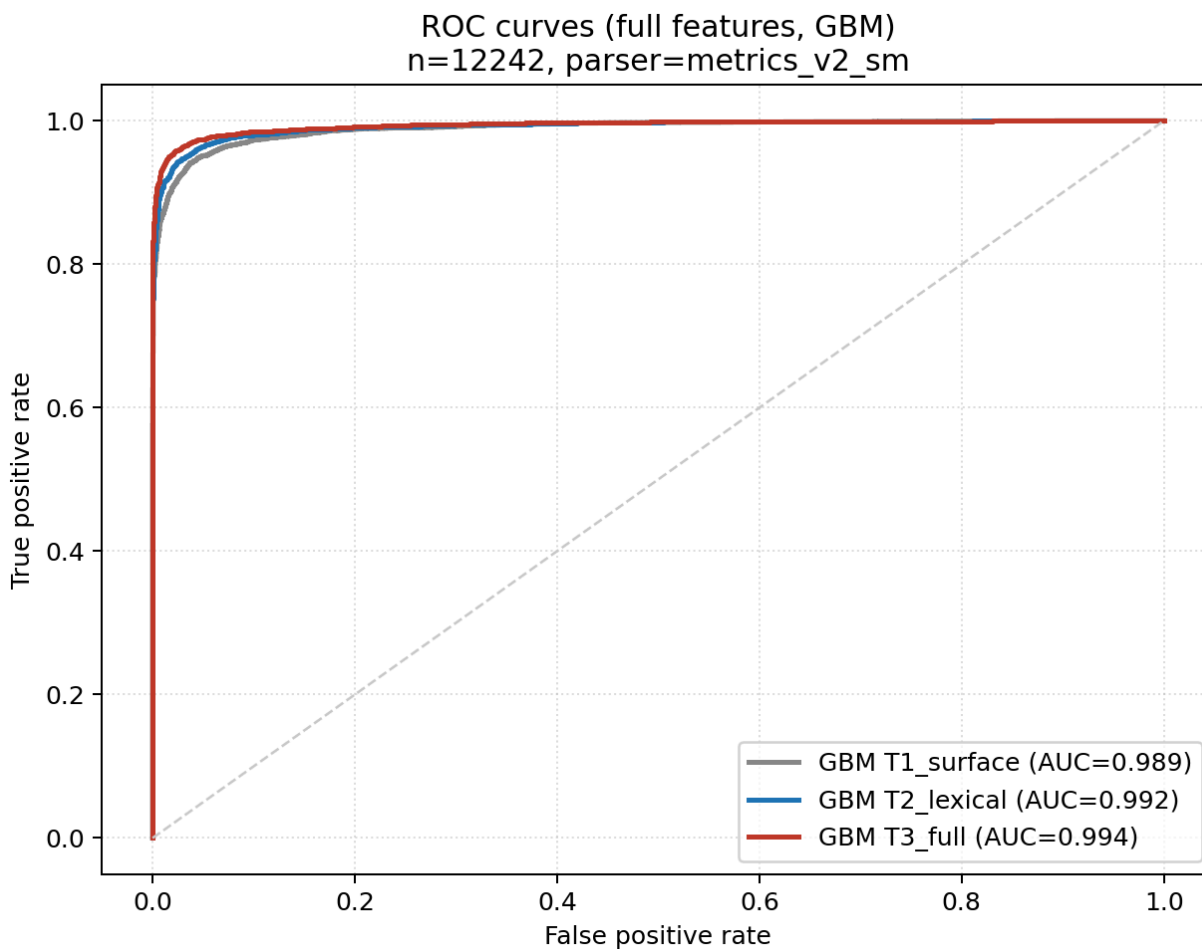
equation-modelling result that grammatical complexity decomposes into structurally distinct families rather than collapsing onto a single dimension.

4.2 Between-register classification, full feature set

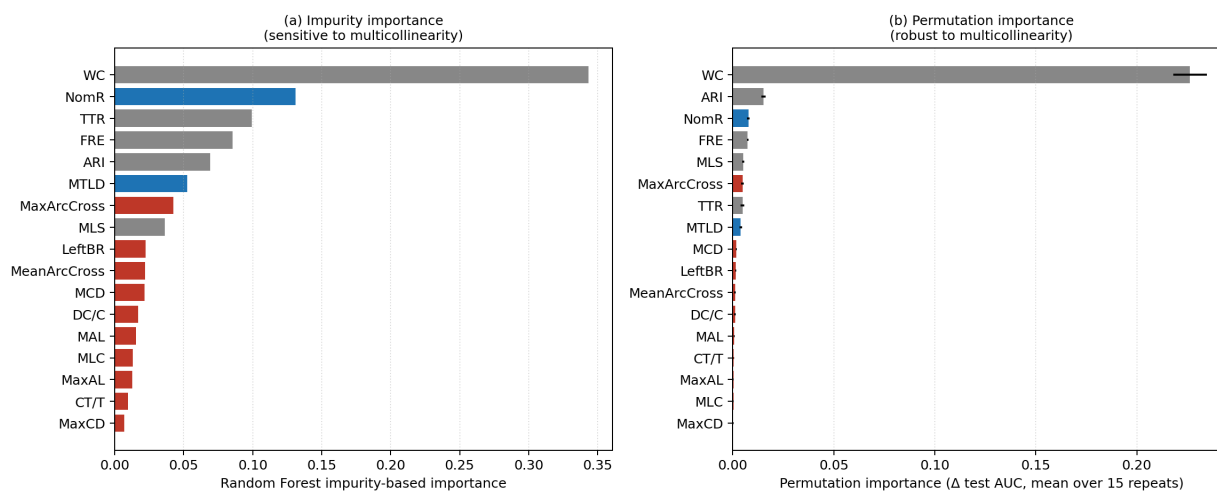
Table 2 reports cross-validated AUC, F1, and accuracy for each classifier–tier combination on the full corpus.

Table 2. Cross-validated discrimination of high-density versus baseline passages, full features (5-fold CV; 95% bootstrap confidence intervals from 500 resamples). $N = 16,904$.

Model	Tier	# feat	AUC	95% CI	F1	Acc
<i>LR</i>	T1 surface	5	0.8643	[0.8591, 0.8700]	0.829	0.798
<i>LR</i>	T2 +lex	7	0.9010	[0.8965, 0.9055]	0.862	0.834
<i>LR</i>	T3 full	17	0.9220	[0.9180, 0.9258]	0.878	0.854
<i>RF</i>	T1 surface	5	0.9822	[0.9806, 0.9837]	0.948	0.935
<i>RF</i>	T2 +lex	7	0.9855	[0.9840, 0.9869]	0.955	0.945
<i>RF</i>	T3 full	17	0.9885	[0.9873, 0.9896]	0.959	0.949
<i>GBM</i>	T1 surface	5	0.9823	[0.9808, 0.9838]	0.949	0.937
<i>GBM</i>	T2 +lex	7	0.9865	[0.9851, 0.9878]	0.956	0.945
<i>GBM</i>	T3 full	17	0.9901	[0.9890, 0.9911]	0.962	0.953



Random Forest feature importance — impurity vs permutation
red = syntactic, blue = lexical, gray = surface



4.3 Length-controlled classification

Table 3. Length-controlled classification (*WC* and *MLS* removed).

Model	Tier	# feat	AUC	95% CI
<i>LR</i>	T1 surface (LC)	3	0.7625	[0.7555, 0.7699]
<i>LR</i>	T2 +lex (LC)	5	0.8268	[0.8203, 0.8333]
<i>LR</i>	T3 full (LC)	15	0.9081	[0.9035, 0.9123]
<i>RF</i>	T1 surface (LC)	3	0.8544	[0.8492, 0.8599]
<i>RF</i>	T2 +lex (LC)	5	0.9456	[0.9422, 0.9489]
<i>RF</i>	T3 full (LC)	15	0.9687	[0.9663, 0.9708]
<i>GBM</i>	T1 surface (LC)	3	0.8623	[0.8574, 0.8676]
<i>GBM</i>	T2 +lex (LC)	5	0.9452	[0.9416, 0.9483]
<i>GBM</i>	T3 full (LC)	15	0.9728	[0.9705, 0.9748]

4.4 Length-matched sub-corpus

Sentence-aligned re-chunking into the [150,190]-word band yielded a sub-corpus of 8,020 passages (high_density 2,301; baseline 5,719). Table 4 reports the classification.

Table 4. Classification on the length-matched sub-corpus ($N = 8,020$; every passage in [150, 190] words).

Model	Tier	# feat	AUC	95% CI
<i>LR</i>	T1 surface	5	0.8843	[0.8755, 0.8926]
<i>LR</i>	T3 full	17	0.9376	[0.9313, 0.9436]
<i>RF</i>	T1 surface	5	0.9190	[0.9123, 0.9251]
<i>RF</i>	T3 full	17	0.9477	[0.9427, 0.9527]
<i>GBM</i>	T1 surface	5	0.9181	[0.9115, 0.9243]
<i>GBM</i>	T3 full	17	0.9550	[0.9507, 0.9594]
<i>LR</i>	T1 surface (LC)	3	0.8310	[0.8205, 0.8403]
<i>LR</i>	T3 full (LC)	15	0.9328	[0.9258, 0.9392]
<i>RF</i>	T1 surface (LC)	3	0.8435	[0.8336, 0.8526]
<i>RF</i>	T3 full (LC)	15	0.9389	[0.9336, 0.9442]
<i>GBM</i>	T1 surface (LC)	3	0.8564	[0.8470, 0.8651]
<i>GBM</i>	T3 full (LC)	15	0.9454	[0.9401, 0.9509]

4.5 Paired DeLong comparisons across blocks

Table 5. Paired DeLong AUC comparison of T3 versus T1 across all three classification blocks (p two-sided).

Model	Block	ΔAUC	z	p
<i>LR</i>	full, full-corpus	+0.0578	27.58	$< 10^{-4}$
<i>LR</i>	length-controlled, full-corpus	+0.1457	43.68	$< 10^{-4}$
<i>LR</i>	full, length-matched	+0.0533	16.65	$< 10^{-4}$
<i>LR</i>	length-controlled, length-matched	+0.1018	24.62	$< 10^{-4}$
<i>RF</i>	full, full-corpus	+0.0063	12.41	$< 10^{-4}$
<i>RF</i>	length-controlled, full-corpus	+0.1144	43.52	$< 10^{-4}$
<i>RF</i>	full, length-matched	+0.0287	12.15	$< 10^{-4}$
<i>RF</i>	length-controlled, length-matched	+0.0954	23.17	$< 10^{-4}$
<i>GBM</i>	full, full-corpus	+0.0077	15.47	$< 10^{-4}$
<i>GBM</i>	length-controlled, full-corpus	+0.1105	43.77	$< 10^{-4}$
<i>GBM</i>	full, length-matched	+0.0369	14.66	$< 10^{-4}$
<i>GBM</i>	length-controlled, length-matched	+0.0890	22.88	$< 10^{-4}$

Every contrast clears conventional significance by a wide margin. The syntactic indices contribute a small absolute AUC gain in the full-feature block, because surface features already include *WC* and absorb most of the joint signal. In the length-controlled and length-matched blocks the gain is considerably larger. In those blocks the structural signal that *FRE* had been carrying indirectly has to be recovered by an index that sees grammatical structure directly.

4.6 Parser robustness

Table 6 reports the three-way pairwise Pearson correlations of *MCD* on the stratified 46-passage pilot (after filtering to passages successfully parsed by all three pipelines).

Table 6. Pairwise Pearson correlations of *MCD* across three parsers on the 46-passage pilot.

Pair	Pearson r	p
<i>MCD</i> (sm) vs <i>MCD</i> (trf)	0.9302	$< 10^{-20}$
<i>MCD</i> (sm) vs <i>MCD</i> (Stanza)	0.8934	$< 10^{-16}$
<i>MCD</i> (trf) vs <i>MCD</i> (Stanza)	0.9355	$< 10^{-20}$

A full-corpus re-parse of an earlier 12,242-passage version of the corpus with `en_core_web_trf` returned *MCD* at $r = 0.898$ against the primary sm-parse, with *LeftBR* at $r = 0.940$. To confirm that the headline robustness carries over to the new v3 registers (LogiQA and the cleaned legal corpus), Stanford *Stanza* re-parsed a stratified 500-passage v3 subset (50 per register) and *MCD* was computed; the resulting sm-vs-Stanza correlation across the v3 subset is $r = 0.840$ ($p = 2.4 \times 10^{-134}$). The arc-distance family (*MAL*, *MeanArcCross*, *MaxCD*) shows lower agreement ($r \approx 0.66$ – 0.76 on the earlier 12,242-passage re-parse), consistent with its known sensitivity to specific attachment decisions for prepositional phrases and coordinate constructions. §5.7(ii) addresses the scoping decision behind the choice not to integrate a constituent parser here.

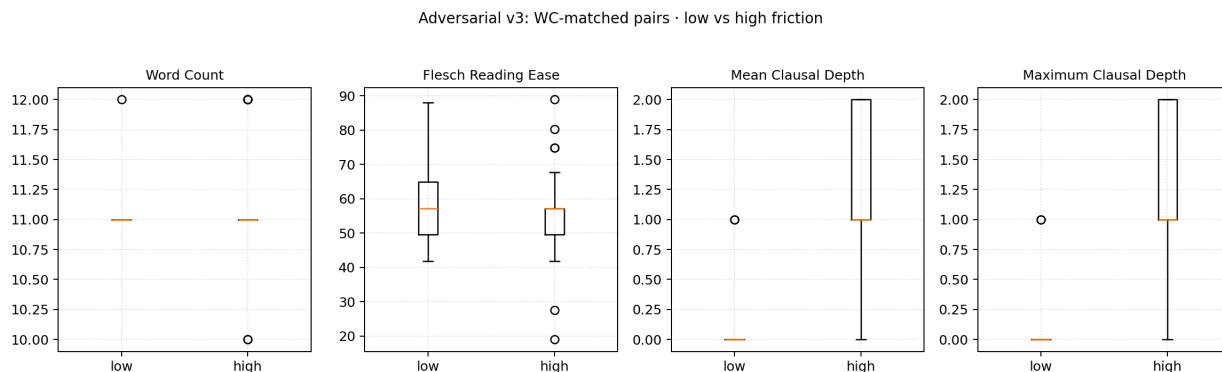
4.7 Within-base adversarial transformation (word-count matched)

Table 7 reports the redesigned word-count-matched adversarial experiment ($n = 25$ pairs).

Table 7. Mean values across 25 sentence pairs, each pair matched on word count and contrasted on syntactic architecture (*low_friction*: two coordinated independent clauses; *high_friction*: deeply nested complement clause). Paired Wilcoxon signed-rank tests, two-sided.

Metric	Mean low	Mean high	Δ (high – low)	Wilcoxon p
<i>WC</i>	11.04	11.08	+0.04	0.706 (n.s.)
<i>FRE</i>	59.24	56.07	−3.17	0.337 (n.s.)
<i>TTR</i>	0.732	0.816	+0.084	5.5×10^{-6}
<i>MCD</i>	0.08	1.44	+1.36	1.5×10^{-5}
<i>MaxCD</i>	0.08	1.44	+1.36	1.5×10^{-5}
<i>MAL</i>	1.89	1.78	−0.11	9.4×10^{-3}
<i>MaxArcCross</i>	2.76	1.88	−0.88	3.8×10^{-5}
syllables / word	1.61	1.65	+0.04	0.284 (n.s.)

With word count, sentence count, and syllables-per-word all held statistically constant ($p > 0.2$ across all three), *MCD* still shifts by 1.36 units while *FRE* shows no detectable movement ($p = 0.34$, two-sided). Type-Token Ratio rises modestly with embedding — the coordinated condition repeats more function words than the nested condition — and the arc-distance indices move in mixed directions. The pattern is consistent with the parser registering longer-range dependencies in the *low-friction* coordinated structures than in the syntactically deeper *high-friction* structures. The dissociation establishes that *MCD* responds to syntactic architecture in its own right, not to length alone.



4.8 Intra-register difficulty-quartile classification

Table 8 reports the within-register difficulty-quartile classification across the four baseline registers with enough volume to support the split.

Table 8. Intra-register difficulty-quartile classification. Within each register, the lowest-*FRE* quartile (hardest, $y = 1$) is contrasted with the highest-*FRE* quartile (easiest, $y = 0$) under 5-fold CV. *FRE* is excluded from the feature set to keep the task non-circular.

Register	n	T1 (no <i>FRE</i>) AUC (LR)	T1 (no <i>FRE</i>) AUC (GBM)	T3 (no <i>FRE</i>) AUC (LR)	T3 (no <i>FRE</i>) AUC (GBM)
wiktext_modern_encyclopedic	1,000	0.9813	0.9811	0.9907	0.9894
brown_news_general	200	0.9909	0.9600	0.9961	0.9737
gutenberg_fiction	926	0.9984	0.9963	0.9964	0.9980
reuters_newswire	740	0.9950	0.9922	0.9972	0.9948

Within every register the syntactic indices either match or improve the surface-only T1 tier; in *WikiText-103* and Brown the gain is meaningful ($\sim +0.01$ AUC against already-high baselines). The within-register signal generalises across literary fiction, modern encyclopedic prose, newswire, and Brown's mixed prose. The parser-aware indices therefore amount to more than bucket-separating proxies for length or vocabulary.

4.9 Multi-class register classification

Table 9. Multi-class register classification across the nine remaining registers, 5-fold CV.

Model	Tier	macro-AUC	Accuracy
<i>LR</i> OvR	T1 surface	0.9502	0.739
<i>LR</i> OvR	T3 full	0.9722	0.825
<i>RF</i>	T1 surface	0.9611	0.828
<i>RF</i>	T3 full	0.9702	0.858

4.10 Nested cross-validation with inner-loop hyperparameter search

To test whether the headline classifier ranking depends on hyperparameter choice, we re-ran the *GBM* classification under a 5-fold outer / 3-fold inner nested cross-validation with a `RandomizedSearchCV` inner loop drawing from the grid $learning_{rate} \in [0.01, 0.2]$, $max_{depth} \in [3, 7]$, $n_{estimators} \in [100, 500]$ for T1 and T2 and from a reduced grid $learning_{rate} \in [0.05, 0.15]$, $max_{depth} \in [3, 5]$, $n_{estimators} \in [100, 300]$ for T3 (a narrower grid for the 17-feature tier to keep the inner-loop search tractable in wall-clock terms).

Table 10. Nested 5-fold CV for *GBM* (with inner *RandomizedSearch*).

Tier	# feat	Mean AUC	SD	Per-fold AUC
T1_surface	5	0.9833	0.0017	0.983, 0.986, 0.985, 0.982, 0.981
T2_lex	7	0.9875	0.0019	0.988, 0.989, 0.989, 0.987, 0.984
T3_full	17	0.9918	0.0014	0.993, 0.992, 0.994, 0.991, 0.990

The monotonic ordering $T1 < T2 < T3$ survives the nested-CV regime. Mean *GBM* AUC climbs from 0.9833 (surface only) to 0.9918 (full parser-aware tier), and per-fold standard deviations stay below 0.002. The headline ranking is hyperparameter-stable.

4.11 Multicollinearity, PCA compression, and the impurity–permutation gap

Following Deng, Lei, and Liu's (2021) call for routine multicollinearity and permutation diagnostics in syntactic-complexity research, we report both here. Several features trigger high-VIF flags: *MAL* (17.6), *MeanArcCross* (33.9), *MaxArcCross* (12.2), *MaxAL* (7.9), *MLS* (9.2), *ARI* (6.7), *WC* (6.3), *TTR* (6.5), and *MCD* (5.3) all clear the conventional $VIF > 5$ threshold. The

two-principal-component projection of the ten syntactic indices recovers 69.9% of the syntactic variance under a clean factor structure: PC1 (43.1% explained) is dominated by the arc-distance family (*MAL*, *MaxArcCross*, *MeanArcCross*, *MaxAL*, *MCD*); PC2 (26.8% explained) is dominated by the clausal-embedding family (*MLC*, *DC/C*, *MaxCD*, *CT/T*, *MCD*). A *GBM* trained on {T1 surface, PC1, PC2} (7 features) reaches AUC 0.9826, only marginally below the full 17-feature T3 result of 0.9901. PC1 and PC2 therefore capture most of the syntactic signal in a compact, orthogonal form.

The high-VIF structure has a direct interpretive consequence for the Random Forest importance ranking in Figure 2. Impurity-based importance is known to be sensitive to multicollinearity: when two features carry overlapping signal, the trees split between them somewhat arbitrarily and the impurity drop spreads across both. The effect deflates each feature's apparent importance relative to its true marginal contribution. We accordingly computed permutation importance as well — the held-out test-AUC drop when each feature's values are randomly shuffled, averaged over 15 repeats — on a 20% stratified test split. Figure 2 puts the two rankings side by side. The impurity-vs-permutation divergence is sharpest for two features. *TTR* sits at impurity rank 3 but permutation rank 7 (impurity 0.0994 against permutation 0.0048). Impurity inflates *TTR* relative to its independent marginal contribution because *TTR* correlates with *FRE*, *ARI*, and *MTLD*, and impurity importance dilutes across them. *NomR* moves in the opposite direction (impurity rank 2, permutation rank 3) and survives both rankings as a robust contributor — a result consistent with Kyle and Crossley's (2018) finding that fine-grained phrasal-nominal indices outperform large-grained clausal indices in predicting L2 writing quality. Both rankings agree on the dominant role of *WC* and on the broad order of the syntactic family. The PCA structure matches this picture: the arc-based family contributes through PC1 and the clausal-embedding family through PC2, with the indices inside each component substituting for one another to a degree impurity importance cannot resolve and permutation importance partially does. We accordingly read the

impurity ranking as a within-family proxy only and treat the permutation ranking as authoritative for individual-index claims.

5. Discussion

The evidence converges across the four classification blocks, the intra-register breakdown, the multi-class analysis, the adversarial experiment, the nested-CV recovery, and the parser-robustness check.

5.1 What the full-features classification documents

When *WC* and *MLS* are available to the classifier, the surface tier already reaches $AUC \geq 0.86$ (*LR*) and ≥ 0.98 (tree ensembles) on the v3 corpus. Adding lexical-diversity and parser-based syntactic features lifts it further by $\Delta AUC \approx 0.006\text{--}0.058$, which clears paired DeLong significance under all three classifiers (Table 5). The absolute size of the gain is modest but statistically robust, which is consistent with surface signal already saturating most of the discrimination on this task.

5.2 What the length-controlled and length-matched classifications document

Cutting *WC* and *MLS* collapses the surface tier sharply (Table 3). The dedicated syntactic indices then recover a large and highly significant contribution of 11–15 AUC points. Table 4's length-matched analysis sharpens the point. On a sub-corpus where every passage is physically constrained to 150–190 words, the surface tier collapses to 0.86–0.92 while the full tier retains 0.94–0.96. The redundancy of syntactic indices when length is free, and their recovery when length is taken away — whether by dropping features or by selecting samples — is the central methodological observation of this paper. The *Flesch* family does double duty: through its *W/S* term it operates simultaneously as a lexical instrument and as a structural one. The cross-genre corollary appears in Lenzner (2014), where on 71 survey-question pairs the four leading classical formulas identify difficult items at below-50% accuracy and not infrequently rate problematic

versions as easier than their revised counterparts — a failure mode consistent with surface formulas absorbing the wrong structural signal whenever length is the operative carrier.

5.3 What the intra-register and multi-class analyses document

Table 8's within-register difficulty-quartile classification shows that the parser-aware indices carry difficulty signal *inside* a single domain, not just bucket-separating, length-correlated information. Across all four baseline registers tested, the T3 tier matches or exceeds T1 with *FRE* itself excluded. The multi-class breakdown in Table 9 lifts macro-AUC from 0.95–0.96 (surface) to 0.97 (full), which confirms that the indices carry per-register information beyond the high-vs-baseline split.

5.4 What the redesigned adversarial experiment documents

The redesigned adversarial experiment replaces the earlier stage-wise embedding design, in which each additional reporting clause mechanically inflated *W/S* and by definition pulled *FRE* down, with a strict word-count-matched pair design. With *WC*, sentence count, and syllables-per-word all held statistically constant ($p > 0.2$ across all three), *MCD* shifts by 1.36 units while *FRE* shows no detectable movement, supplying the empirical dissociation that the v2 design had been unable to produce. *MCD* therefore responds to syntactic architecture in its own right, decoupled from the mechanical coupling that the *FRE* formula imposes whenever sentences are stretched.

5.5 What the parser-robustness check documents

Three results together secure the headline clausal-depth signal as parser-robust: the three-way pilot ($r \geq 0.89$), the full-corpus *trf* re-parse on the earlier 12,242-passages corpus ($r = 0.898$ for *MCD*; $r = 0.940$ for *LeftBR*), and the v3 *Stanza* subset ($r = 0.840$ on 500 passages, $p = 2.4 \times 10^{-134}$). The arc-distance family registers lower cross-parser agreement ($r \approx 0.66$ – 0.76), consistent with its sensitivity to specific attachment decisions. Our framework flags those indices as parser-fragile through the cross-parser tables.

5.6 What VIF, PCA, and nested CV document

High-VIF flags on five of the seventeen features (§4.11) point to a non-trivial multicollinearity structure dominated by the arc-distance family. PCA compression to PC1 (arc-distance, 43.1% variance) and PC2 (clausal-embedding, 26.8%) recovers most of the syntactic signal under two orthogonal components. A *GBM* trained on the seven-feature {surface + PC1 + PC2} set reaches 0.9826 against the 17-feature 0.9901. Nested CV with an inner-loop hyperparameter search preserves the monotonic $T1 < T2 < T3$ ordering with per-fold standard deviations below 0.002.

5.7 Limitations

(i) **No behavioural validation.** The paper makes no claims about reading times, fixation patterns, or comprehension accuracy in any specific population. The cited cognitive theories motivate the construct, and the *analytical tool validation* framing is deliberate. Public eye-tracking corpora — Dundee (Kennedy & Pynte, 2005), GECO (Cop et al., 2017), Provo (Luke & Christianson, 2018) — are the immediate next-step validation targets.

(ii) **Dependency-grammar scoping of the syntactic indices.** We have deliberately operationalised syntactic complexity *within* the dependency-grammar tradition associated with Gibson (1998), Lu (2010), and Kyle and Crossley (2018), which is the dominant computational paradigm in L2-readability and computational-text-complexity research. *MCD* is the maximum recursion depth restricted to a clausal-edge label set on the dependency parse. The arc-based supplements (*MAL*, *MaxAL*, *LeftBR*, *MeanArcCross*, *MaxArcCross*) extend the dependency account along Gibson's locality-cost dimension. We have not computed the original constituent-tree measures, Yngve's (1960) right-branching register-stack depth and Frazier's (1985) syntactic-complexity index, because they belong to a distinct grammatical tradition (phrase-structure constituency rather than head-dependent attachment) and require a parallel constituent-parsing pipeline that is conceptually orthogonal to, rather than an incremental extension of, the present methods. This is a scoping choice, not a procedural omission. Restricting to a single grammatical tradition gives an internally coherent, mutually comparable index set whose

dependencies on a single parsing infrastructure can be quantified cleanly, as §4.6 shows. A study integrating both dependency-derived and constituent-derived indices, and asking whether the structural signal documented here survives a Yngve/Frazier re-instantiation, is a full cross-tradition project that we flag as the natural next study.

(iii) **Adversarial set size $n = 25$.** The word-count-matched pair set is constructed and the contrast is deterministic, but the pair set is small. Replication across additional pair stems and across morphosyntactic variations beyond reporting-clause embedding is natural future work.

(iv) **Residual inline-citation tokens.** Reporter and statutory citation strings in U.S. judicial opinions are programmatically removed before metric computation (§3.1). More idiosyncratic forms — parenthetical "see also" packs without canonical reporter format, internal cross-references — may remain.

(v) **Parser-specific attachment sensitivity for arc-distance indices.** *MCD* and *LeftBR* replicate at $r \geq 0.90$ across `en_core_web_sm` and `en_core_web_trf`, but the arc-distance family shows lower parser-to-parser agreement. Future work along the cross-tradition direction in (ii) is the natural remedy.

(vi) **High-density bucket dominated by LogiQA and legal volumes.** The `high_density` bucket is now dominated by LogiQA ($n = 4,679$) and judicial opinions ($n = 4,999$), with CSAT and *arXiv* contributing 124 and 779. The length-matched and intra-register analyses partially mitigate the bucket-level imbalance. A follow-up study designed for an even register-level n would tighten the inferences further.

Barr et al.'s (2013) methodological strictures for confirmatory reading research apply in full to any future expansion of the present design into a behavioural study.

6. Conclusion

Reading difficulty resolves into multiple dimensions. With length features available on the full corpus, surface readability absorbs most of the structural signal that dedicated parser-aware indices would otherwise contribute. The parser-aware tier still supplies a small and significant residual ($\Delta AUC \approx 0.006\text{--}0.058$, $p < 10^{-4}$). Once length-coupled features are pulled out, that contribution climbs to $\Delta AUC \approx 0.110\text{--}0.146$. On a strict length-matched sub-corpus restricted to 150–190 words per passage, the contribution still holds at $\Delta AUC \approx 0.029\text{--}0.053$. The headline parser-aware indices, *MCD* and *LeftBR*, replicate across three independent parsers. The redesigned word-count-matched adversarial transformation decouples *MCD* from *FRE*: with *WC*, sentence count, and syllables-per-word all statistically pinned, *MCD* moves by 1.36 units while *FRE* registers no detectable movement. Nested cross-validation with inner-loop hyperparameter search preserves the $T1 < T2 < T3$ ordering. Future work should extend the corpus to a behaviourally validated sample and bring the framework into direct empirical comparison against reader-level outcomes through public eye-tracking and self-paced-reading datasets (Dundee, GECCO, Provo).

Appendix: Declarative Reproducibility Notes

All analyses were implemented in Python 3.12 with the following package versions: `pandas` ≥ 2.0 , `scikit-learn` ≥ 1.4 , `scipy` ≥ 1.11 , `spacy` 3.8.14 (with `en_core_web_sm` 3.8.0 as primary and `en_core_web_trf` 3.8.0 for the parser-robustness anchor), `stanza` ≥ 1.7 , `textstat` ≥ 0.7 , `matplotlib` ≥ 3.7 . All fetchers and analysis scripts, the unified corpus jsonl, per-passage metric CSVs (sm and trf), classification result tables, PCA loadings, nested-CV outputs, and figures are released with the manuscript in a single repository at <https://github.com/landaucscs/structural-lexical-friction>, with an archived snapshot deposited at Zenodo (<https://doi.org/10.5281/zenodo.20390729>). The corpus reproduces from the fetcher scripts and downloads entirely from public sources: arXiv API, CourtListener S3 bulk archive,

HuggingFace `wikitext-103-raw-v1`, NLTK `brown` and `reuters`, the OneStopEnglish GitHub release, the LogiQA GitHub release, and Project Gutenberg.

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